

a. Monocot



b. Eudicot

Figure 23.21 Flower diversity. Regardless of size and shape, flowers, such as this (**a**) monocot and (**b**) eudicot, share certain features.
(a): Ed Reschke/Stone/Getty Images; (b): Steven P. Lynch

structure, a saclike container called the anther. Pollen grains develop from microspores produced within the anther.

- At the very center of a flower is the **carpel,** a vaselike structure with three major regions: the stigma, an enlarged, sticky knob; the style, a slender stalk; and the **ovary,** an enlarged base that encloses one or more ovules. The ovule becomes the seed, and the ovary becomes the fruit. Fruit is often instrumental in the distribution of seeds.

Note that not all flowers have all these parts (Table 23.3). A flower is said to be *complete* if it has all four parts; otherwise, it is *incomplete.*

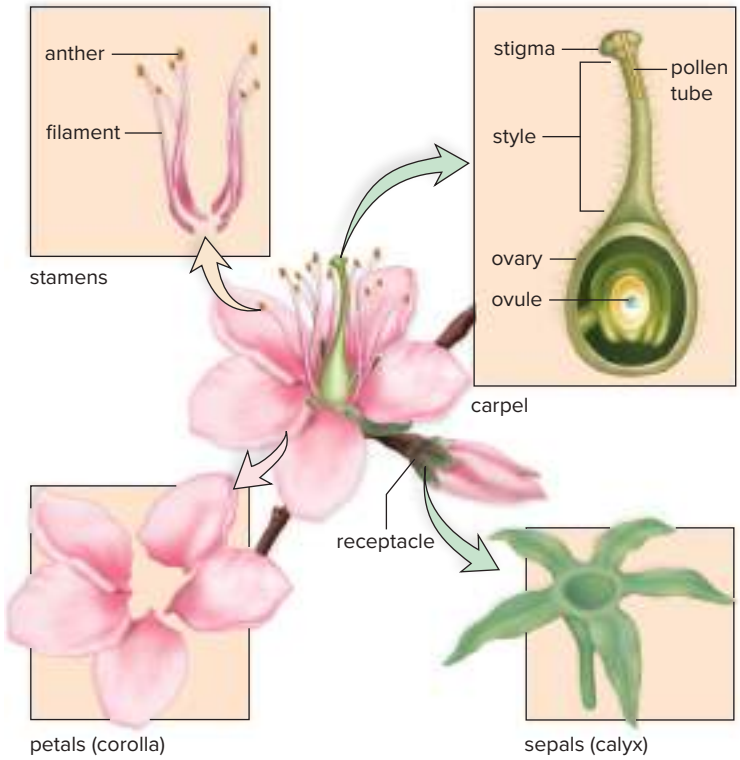


Figure 23.22 Generalized flower. A flower has four main parts: sepals, petals, stamens, and carpels. A stamen has an anther and filament. A carpel has a stigma, style, and ovary. An ovary contains ovules.

Table 23.3 Other Flower Terminology

Term	Description of Flower
Complete	All four parts (sepals, petals, stamens, and carpels) present
Incomplete	Lacks one or more of the four parts
Perfect	Has both stamens and (a) carpel(s)
Imperfect	Has stamens or (a) carpel(s), but not both
Inflorescence	A cluster of flowers
Composite	Appears to be a single flower but consists of a group of tiny flowers

Flowering Plant Life Cycle

Figure 23.23 depicts the life cycle of a typical flowering plant. Like the gymnosperms, flowering plants are heterosporous, producing two types of spores. A megaspore located in an ovule within an ovary of a carpel develops into an egg-bearing female gametophyte, called the embryo sac. In most angiosperms, the embryo sac has seven cells; one of these is an egg, and another contains two polar nuclei. They are called the polar nuclei because they came from opposite ends of the embryo sac.

Microspores, produced within anthers, become pollen grains that, when mature, are male gametophytes with sperm. The mature male gametophyte consists of only three cells: the tube cell and two sperm cells.

During pollination, pollen is transported from the anther to the stigma of a carpel. Here, the tube cell produces a pollen tube that

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carries the two sperm to the micropyle (small opening) of an ovule. Flowering plants undergo **double fertilization:** One sperm unites with an egg, forming a diploid zygote, and the other unites with polar nuclei, forming a triploid endosperm nucleus (see Section 27.1).

Ultimately, the ovule becomes a seed that contains the embryo (the sporophyte of the next generation) and stored food enclosed within a seed coat (see Fig. 23.2*b*). Endosperm in some seeds is absorbed by the cotyledons, whereas, in other seeds, endosperm is digested as the seed matures.

A **fruit** is derived from an ovary, and in some instances it is an accessory part of the flower. Some fruits, such as apples and tomatoes, provide a moist, fleshy covering; other fruits, such as pea pods and acorns, provide a dry covering for seeds.

Flowers and Diversification

Flowers are involved in the production and development of spores, gametophytes, gametes, and embryos enclosed within seeds. Successful completion of sexual reproduction in angiosperms requires the effective dispersal of pollen and then seeds. The various ways pollen and seeds can be dispersed have resulted in many different types of flowers (see the Evolution feature, "Plants and Their Pollinators," in Section 27.1).

Wind-pollinated flowers are usually not showy, whereas many insect- and bird-pollinated flowers are colorful. Night-blooming flowers attract nocturnal mammals or insects; these flowers are usually aromatic and white or cream-colored.

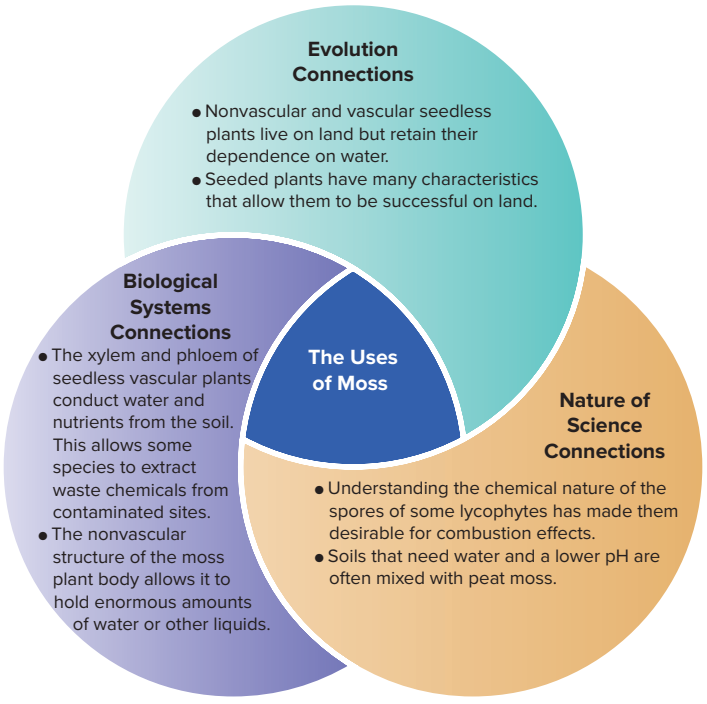
Although some flowers disperse their pollen by wind, many are adapted to attract specific pollinators, such as bees, wasps, flies, butterflies, moths, and even bats, that carry pollen from one flower to another flower of the same type. For example, glands in the region of the ovary produce nectar, a nutrient that is gathered by pollinators as they go from flower to flower. Bee-pollinated flowers are usually blue or yellow and have ultraviolet shadings that lead the pollinator to the location of the nectar. The mouthparts of bees are fused into a long tube, which is able to obtain nectar from the base of the flower. In another example, instead of beautiful colors and tempting nectar, some species of *Ophyrys* orchids use "sexual mimicry" to attract their wasp pollinators. The orchid flowers look like a female wasp and emit pheromones to attract male wasps. The males will engage in "pseudocopulation" with the "female" and end up with pollen attached to its head. Frustrated, the male wasp leaves the flower and attempts to mate with another flower. The male then deposits pollen to the second flower, thereby completing the cross-pollination that the orchid intended.

The fruits of flowers protect and aid in the dispersal of seeds. Dispersal occurs when seeds are transported by wind, gravity, water, and animals to another location. Fleshy fruits may be eaten by animals, which transport the seeds to a new location and then deposit them when they defecate. Because animals live in particular habitats and/or have particular migration patterns, they are apt to deliver the fruit-enclosed seeds to a suitable location for seed germination (when the embryo begins to grow again) and development of the plant.

Check Your Progress 23.5

1. List the life cycle changes that have enabled pines to better adapt to life on land.
2. Compare and contrast the four types of gymnosperms.
3. List the functions of the key structures required for angiosperm reproduction.

Connecting *the* Concepts



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